Hojun Yang

M.S. Researcher - Robotics Field Autonomy & Geospatial Data Engineering
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RESEARCH INTERESTS

Field robotics and automation for mining and hazardous environments: GPS-denied underground exploration and navigation; mobile manipulation (robot arm) and robot learning for mining-equipment automation; UAV-based spatial AI and photogrammetry; and planetary surface robotics and in-situ resource utilization (ISRU).

EDUCATION

M.S. in Energy Resources and Geosystems Engineering Mar 2025 – Expected Feb 2027

Sejong University, Seoul, South Korea · SMART-X Lab · Advisor: Prof. Yosoon Choi
Thesis: Developing an AI-based UAV photogrammetry pipeline that removes mining equipment from image datasets and improves Digital Elevation Model (DEM) reconstruction accuracy in open-pit mines
Coursework: Microwave Remote Sensing, Geostatistics, Electro-Chemical Engineering, Energy Economics
GPA: 4.30 / 4.5

B.S. in Energy Resources Engineering Mar 2020 – Feb 2025

Pukyong National University, Busan, South Korea · Advisor: Prof. Yosoon Choi
Thesis: Autonomous Navigation and Automated Logistics System Using Smart Robots
Coursework: Machine Learning, Robotics, Mineral Processing, Geology, Rock Mechanics, Geothermal, Smart Grid
GPA: 4.34 / 4.5 (Class rank: 1 of 13)

JOURNAL PUBLICATIONS

Yang, H., Ahn, T., & Choi, Y. (2026). Development of a YOLO-based detection model for key visual defect elements of tower cranes using UAV images. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

Yang, H., & Choi, Y. (2026). Generation of vehicle-free digital elevation models in open-pit mines using UAV image inpainting. *Measurement, 260*, 119902. **[SCIE]**

Yang, H., Kim, S., & Choi, Y. (2025). Development of a drone- and AI-based framework for exterior inspection and damage classification of tower cranes. *Tunnel and Underground Space*.

Yang, H., Kim, S., & Choi, Y. (2025). Review of UAV–AI-based automated visual inspection technologies and their applicability to high-risk construction machinery. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

Jang, M., **Yang, H.**, & Choi, Y. (2025). Development of augmented reality-based educational content for mine workers. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

Choi, S., **Yang, H.**, Kim, Y., & Choi, Y. (2025). Router-drop-based robot teleoperation system for unconnected underground mines. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

Yang, H., & Choi, Y. (2025). Analysis of wheel slip and path errors based on steering mechanisms of a small autonomous robot in underground rough environments. *Tunnel and Underground Space*.

CONFERENCE PRESENTATIONS

Yang, H., & Choi, Y. UAV image inpainting for vehicle-free DEM generation in open-pit mines. *SME Annual Conference & Expo (MINEXCHANGE 2026)*, Salt Lake City, UT, USA, Feb 2026. (Poster)

Yang, H., et al. Scaled collaborative-manipulator platform for robot-learning-based autonomous mining equipment. *KSMER Spring Conference*, Gyeongju, Korea, 2026. (Poster)

Yang, H., & Choi, Y. UAV image inpainting for open-pit mine DEM reconstruction. *KSMER Fall Conference*, Busan, Korea, 2025. (Oral) **- Best Paper Presentation Award**

Yang, H., & Choi, Y. Removing heavy equipment from open-pit mine DEMs via deep inpainting. *KSMER Spring Conference*, Jeju, Korea, 2025. (Oral)

Yang, H., & Choi, Y. Wheel slip and path error analysis of steering mechanisms in underground rough terrain. *KSMER Fall Joint Conference*, High1 Resort, Gangwon, Korea, 2024. (Oral)

AWARDS & HONORS

Research Assistantship, Sejong University	2025-2026
Future Talent Scholarship, Pukyong National University	2024
Academic Excellence Scholarship, Pukyong National University	2020, 2023-2024
Best Paper Presentation Award, KSMER Fall Conference	2025
Gold Prize, Mine-Tech Festa (KOMIR) - GeoInpainter	2025
President's Award, Korean Society of Mineral and Energy Resources Engineers	2025
Commendation, Korean Association of Geographic Information Studies	2025
Honorable Mention, LINE Capstone (Pukyong National University)	2024
Grand Prize, Mine-Tech Festa (KOMIR) - MeshRover Connect	2024
Excellence Award, Busan Future Scientist Award (Busan Sci. & Tech. Council)	2024
Excellence Award, Geoscience Data Utilization Competition (KIGAM)	2023
Excellence Award, Mine-Tech Festa (KOMIR) - AR safety content	2023

RESEARCH EXPERIENCE

Graduate Researcher - SMART-X Lab, Sejong University 2026-2027

- **GeoInpainter - Vehicle-free Digital Elevation Model (DEM) generation:** built an automated pipeline (YOLOv8 detection → SAM masking → LaMa/MAT/DeepFillv2 inpainting → SfM) that removes heavy equipment from UAV imagery before 3D reconstruction, cutting elevation error from MAE 1.276 m to 0.111 m for a loader. First-author SCIE paper (*Measurement, 2026*).
- **Mobile manipulation via vision-language-action models:** trained a UGV + 6-DoF collaborative arm for autonomous pick-and-place; forward-kinematics-based dataset auditing raised success to 90% (95-episode dataset). Current M.S. thesis direction.
- **MeshRover Connect - Underground teleoperation:** co-developed a quadruped that autonomously drops mesh-network relay nodes to extend teleoperation in communication-denied mines; demonstrated in an active limestone mine. Grand Prize, Mine-Tech Festa 2024.
- **Steering & wheel-slip analysis:** compared Ackermann, differential, and tracked steering on rough mine terrain using laser-grid ground truth plus simulation (best path RMSE 0.2311 m). First-author paper.
- **Drone + AI inspection of high-risk machinery:** three-stage parallel YOLO architecture (ROI → part segmentation → defect detection) for UAV-based tower-crane inspection (defect mAP 0.937); South Korea government-funded.

RESEARCH PROJECT

Development of a Robot Operating System (ROS) Extension Module for 3D Autonomous Navigation in Underground Rough-Terrain Environments (nuclear disaster site, lunar lava cave, abandoned mines) 2023-2027

- Developed a router-drop-based robot teleoperation system for unconnected underground mines
- Utilized a thermal imaging camera to locate victims and sends a signal upon detection
- The system allows one potentially to investigate abandoned mines, nuclear disaster and lunar lava caves

Development of Safety Assessment and Management Technology for High-Risk Construction Machinery 2025-

- Proposed a systematic framework with ten major structural components of tower failures and seven major categories for defining inspection items and classifying damage types
- Established quantitative criteria for each category to standardize AI training datasets and support the development of automated damage detection models
- The proposed framework provides a technical foundation for constructing smart inspection platforms and advancing automation technologies for the safety assessment of construction equipment

Development of a Compact Planetary Rover System for the Korea Space Rover Challenge (KSRC) 2026
2026-Organized by the Korea AeroSpace Administration (KASA)

- Designed an automated planetary-surface robot engineered for stable locomotion across mixed regolith conditions, including salt-rich and sandstone terrains.

TECHNICAL SKILLS

Robotics & autonomy: ROS (Noetic); LiDAR SLAM (FAST-LIO/FAST-LIO2, LeGO-LOAM); elevation mapping & traversability analysis; MoveIt; behavior trees; Gazebo simulation

Perception & deep learning: YOLOv8/v11; SAM; image inpainting (LaMa, MAT, DeepFillv2); OpenCV; PyTorch

Robot learning: imitation learning; ACT; SmolVLA / LeRobot

Photogrammetry & GIS: UAV mapping; Structure-from-Motion (Pix4D, Metashape); DEM & point-cloud analysis; GeoAI; QGIS

Hardware & platforms: AgileX Scout Mini / LIMO; Unitree Go1; MyCobot Pro 600; Velodyne VLP-16; Intel RealSense D435i/D455; Xsens IMU; NVIDIA Jetson AGX Orin

Programming: Python; Linux/shell

LANGUAGES

Korean (native); English (research & professional working proficiency)